

PostgreSQL in Amazon Linux (Putty)

1) sudo dnf install postgresql15 postgresql15-server -y

```
[ec2-user@ip-172-31-31-51 ~]$ sudo dnf install postgresql15 postgresql15-server -y
Last metadata expiration check: 0:03:24 ago on Tue Sep 22 21:34:36 2026.
Dependencies resolved.
=====
Package                                Architecture      Version           Repository        Size
=====
Installing:
postgresql15                          x86_64            15.19-1.amzn2023.0.1  amazonlinux      1.7 M
postgresql15-server                   x86_64            15.19-1.amzn2023.0.1  amazonlinux      6.3 M
Installing dependencies:
libicu                                x86_64            67.1-7.amzn2023.0.4   amazonlinux      9.6 M
postgresql15-private-libs             x86_64            15.19-1.amzn2023.0.1  amazonlinux      147 k
=====
Transaction Summary
=====
Install 4 Packages

Total download size: 18 M
Installed size: 66 M
Downloading Packages:
(1/4): postgresql15-15.19-1.amzn2023.0.1.x86_64.rpm                33 MB/s | 1.7 MB  00:00
(2/4): postgresql15-private-libs-15.19-1.amzn2023.0.1.x86_64.rpm   2.6 MB/s | 147 kB  00:00
(3/4): libicu-67.1-7.amzn2023.0.4.x86_64.rpm                      50 MB/s | 9.6 MB  00:00
(4/4): postgresql15-server-15.19-1.amzn2023.0.1.x86_64.rpm        32 MB/s | 6.3 MB  00:00
-----
Total                                                                56 MB/s | 18 MB  00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : 1/1
  Installing     : postgresql15-private-libs-15.19-1.amzn2023.0.1.x86_64 1/4
  Installing     : postgresql15-15.19-1.amzn2023.0.1.x86_64             2/4
  Installing     : libicu-67.1-7.amzn2023.0.4.x86_64                   3/4
  Running scriptlet: postgresql15-server-15.19-1.amzn2023.0.1.x86_64     4/4
  Installing     : postgresql15-server-15.19-1.amzn2023.0.1.x86_64     4/4
  Running scriptlet: postgresql15-server-15.19-1.amzn2023.0.1.x86_64     4/4
  Verifying      : libicu-67.1-7.amzn2023.0.4.x86_64                   1/4
  Verifying      : postgresql15-15.19-1.amzn2023.0.1.x86_64            2/4
  Verifying      : postgresql15-private-libs-15.19-1.amzn2023.0.1.x86_64 3/4
  Verifying      : postgresql15-server-15.19-1.amzn2023.0.1.x86_64     4/4

Installed:
libicu-67.1-7.amzn2023.0.4.x86_64  postgresql15-15.19-1.amzn2023.0.1.x86_64  postgresql15-private-libs-15.19-1.amzn2023.0.1.x86_64  postgresql15-server-15.19-1.amzn2023.0.1.x86_64

Complete!
[ec2-user@ip-172-31-31-51 ~]$
```

2) sudo postgresql-setup --initdb

```
[ec2-user@ip-172-31-31-51 ~]$ sudo postgresql-setup --initdb
* Initializing database in '/var/lib/pgsql/data'
* Initialized, logs are in /var/lib/pgsql/initdb_postgresql.log
```

3) sudo systemctl start postgresql

4) sudo systemctl enable postgresql

```
[ec2-user@ip-172-31-31-51 ~]$ sudo systemctl start postgresql
[ec2-user@ip-172-31-31-51 ~]$ sudo systemctl enable postgresql
Created symlink /etc/systemd/system/multi-user.target.wants/postgresql.service →
/usr/lib/systemd/system/postgresql.service.
```

5) sudo systemctl status postgresql

```
[ec2-user@ip-172-31-31-51 ~]$ sudo systemctl status postgresql
● postgresql.service - PostgreSQL database server
   Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-09-22 21:43:15 UTC; 1min 31s ago
     Main PID: 27434 (postgres)
       Tasks: 7 (limit: 1059)
      Memory: 19.3M
         CPU: 101ms
    CGroup: /system.slice/postgresql.service
            └─27434 /usr/bin/postgres -D /var/lib/pgsql/data
              └─27435 "postgres: logger "
                └─27436 "postgres: checkpointer "
                  └─27437 "postgres: background writer "
                    └─27439 "postgres: walwriter "
                      └─27440 "postgres: autovacuum launcher "
                        └─27441 "postgres: logical replication launcher "
```

Sep 22 21:43:15 ip-172-31-31-51.ec2.internal systemd[1]: Starting postgresql.service - PostgreSQL database server...

Sep 22 21:43:15 ip-172-31-31-51.ec2.internal postgres[27434]: 2026-09-22 21:43:15.629 UTC [27434] LOG: redirecting log output to logging collector process

Sep 22 21:43:15 ip-172-31-31-51.ec2.internal postgres[27434]: 2026-09-22 21:43:15.629 UTC [27434] HINT: Future log output will appear in directory "log".

Sep 22 21:43:15 ip-172-31-31-51.ec2.internal systemd[1]: Started postgresql.service - PostgreSQL database server.

lines 1-20/20 (END) ...skipping...

```
● postgresql.service - PostgreSQL database server
   Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-09-22 21:43:15 UTC; 1min 31s ago
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                    └─27439 "postgres: walwriter "
                      └─27440 "postgres: autovacuum launcher "
                        └─27441 "postgres: logical replication launcher "
```

Press Shift+q

sudo su - postgres

psql

```
[ec2-user@ip-172-31-31-51 ~]$ sudo su - postgres
Last login: Tue Sep 22 22:25:10 UTC 2026 on pts/1
[postgres@ip-172-31-31-51 ~]$ psql
psql (15.19)
Type "help" for help.

postgres=# help
You are using psql, the command-line interface to PostgreSQL.
Type:  \copyright for distribution terms
       \h for help with SQL commands
       \? for help with psql commands
       \g or terminate with semicolon to execute query
       \q to quit
postgres=#
```

Create TABLE for CARS

CREATE TABLE cars (brand VARCHAR (255), model VARCHAR (255), INT);

```
postgres=# CREATE TABLE cars (  
postgres(# brand VARCHAR (255),  
postgres(# model VARCHAR (255),  
postgres(# year INT  
postgres(# );  
CREATE TABLE  
postgres=# SELECT * FROM cars;  
  brand | model | year  
-----+-----+-----  
(0 rows)
```

Insert multiple rows

```
postgres=# INSERT INTO cars (brand,model,year)  
postgres-# VALUES  
postgres-# ('VOLVO', 'P1800', 1968),  
postgres-# ('BMW', 'M1', 1978),  
postgres-# ('TOYOTA' 'CELICA', 1975);  
ERROR:  syntax error at or near "'CELICA'"  
LINE 5: ('TOYOTA' 'CELICA', 1975);  
                ^  
  
postgres=# INSERT INTO cars (brand,model,year)  
VALUES  
( 'VOLVO', 'P1800', 1968),  
( 'BMW', 'M1', 1978),  
( 'TOYOTA', 'CELICA', 1975);  
INSERT 0 3  
postgres=# SELECT * FROM CARS;  
  brand | model | year  
-----+-----+-----  
VOLVO  | P1800 | 1968  
BMW    | M1    | 1978  
TOYOTA | CELICA | 1975  
(3 rows)
```

Insert another table

```
postgres=# INSERT INTO cars (brand,model,year)
VALUES
('FORD', 'MUSTANG', 1964);
INSERT 0 1
postgres=# SELECT * FROM CARS;
 brand | model | year
-----+-----+-----
VOLVO  | P1800 | 1968
BMW    | M1    | 1978
TOYOTA | CELICA | 1975
FORD   | MUSTANG | 1964
(4 rows)
```

ADD COLUMN

```
postgres=# alter table cars add color varchar(255);
ALTER TABLE
postgres=# select * from cars;
 brand | model | year | color
-----+-----+-----+-----
VOLVO  | P1800 | 1968 | 
BMW    | M1    | 1978 | 
TOYOTA | CELICA | 1975 | 
FORD   | MUSTANG | 1964 | 
(4 rows)
```

UPDATE THE NEW TABLE

```
postgres=# update cars set color='Blue'
where brand='VOLVO';
```

```
UPDATE 1
```

```
postgres=# select * from cars;
 brand | model | year | color
-----+-----+-----+-----
  BMW  | M1    | 1978 | 
  TOYOTA | CELICA | 1975 | 
  FORD  | MUSTANG | 1964 | 
  VOLVO | P1800  | 1968 | Blue
(4 rows)
```

```
postgres=# update cars set color='Red'
where brand='FORD';
```

```
UPDATE 1
```

```
postgres=# update cars set color='WHITE'
where brand='TOYOTA';
```

```
UPDATE 1
```

```
postgres=# update cars set color='Red'
where brand='FORD';
```

```
UPDATE 1
```

```
postgres=# select * from cars;
 brand | model | year | color
-----+-----+-----+-----
  BMW  | M1    | 1978 | 
  VOLVO | P1800  | 1968 | Blue
  TOYOTA | CELICA | 1975 | WHITE
  FORD  | MUSTANG | 1964 | Red
(4 rows)
```

```
postgres=# update cars set color='BLACK'
where brand='BMW';
```

```
UPDATE 1
```

```
postgres=# select * from cars;
 brand | model | year | color
-----+-----+-----+-----
  VOLVO | P1800  | 1968 | Blue
  TOYOTA | CELICA | 1975 | WHITE
  FORD  | MUSTANG | 1964 | Red
  BMW  | M1    | 1978 | BLACK
(4 rows)
```

DROP COLUMN (Delete the coloumn)

```
postgres=# alter table cars drop column color;
ALTER TABLE
postgres=# select * cars;
ERROR:  syntax error at or near "cars"
LINE 1: select * cars;
              ^

postgres=# select * from cars;
 brand | model | year
-----+-----+-----
VOLVO  | P1800 | 1968
TOYOTA | CELICA | 1975
FORD   | MUSTANG | 1964
BMW    | M1     | 1978
(4 rows)
```

To DELETE the Row

```
postgres=# delete from cars where brand='VOLVO';
DELETE 1
postgres=# select * from cars;
 brand | model | year
-----+-----+-----
TOYOTA | CELICA | 1975
FORD   | MUSTANG | 1964
BMW    | M1     | 1978
(3 rows)
```

Delete the Entire Row

```
postgres=# delete from cars;
DELETE 3
postgres=# select * from cars;
 brand | model | year
-----+-----+-----
(0 rows)

postgres=# truncate table cars;
TRUNCATE TABLE
postgres=# select * from cars;
 brand | model | year
-----+-----+-----
(0 rows)
```

The DROP TABLE Statement

```
postgres=# drop table cars;
DROP TABLE
postgres=# select * from cars;
ERROR:  relation "cars" does not exist
LINE 1: select * from cars;
                        ^
```